

EE 102 Week 4, Lecture 1 (Fall 2026)

Date: September 21, 2026

Contents

1	Goals for today	2
2	Review: Signals and Systems	2
2.1	Example: A matrix as a system	2
2.2	Example: An amplifier as a system	3
2.3	Example: A differential equation as a system	3
2.4	Example: A frequency modulation system	3
3	LTI Output	7
3.1	Example: LTI system response	7
3.2	Example: LTI system properties to compute system response	7
4	Representing signals using impulses	8
4.1	Discrete-time step using impulse	9
4.2	Impulse as a sampler in continuous-time	9
4.3	Example: Representing a sinusoidal signal using impulses	10
4.4	General representation of signals using impulses	11
5	General output of an LTI system using impulse response	12
5.1	Impulse response of an LTI system	12